Supplementary Material for "Stable and Efficient Anomaly Detection in Streaming Graphs"

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Section S1 contains the proofs omitted from Sections 4 and 5, and Section S2 contains additional figures. Theorem, proposition, corollary, definition, equation, table, and section numbers refer to the main paper; figures in this document are numbered S1, S2.

S1 OMITTED PROOFS

Proof of Proposition 4.4. Between two consecutive events of v, coordinate i of the sketch is multiplied by

$$e^{-\mu_i g_{v,k}}.$$

These factors compose to

$$e^{-\mu_i (t-t_k)}$$

from time t_k to time t. Unrolling Eq. (4) therefore gives

$$s_{v,i}(t) = \sum_{\substack{k:\, t_k \le t \\ v \in e_k}} e^{-\mu_i (t-t_k)}\, b_i\, \psi_v(k).$$

Taking the inner product with $\mathbf{w}$ and exchanging the two sums yields Eq. (5). □

Proof of Corollary 4.5. Since $\Delta_v \in [0,1]$ and $\alpha > 0$, every increment is nonnegative. Thus,

$$\mathbf{w}^\top \mathbf{s}_v \ge 0.$$

By Corollary 5.2 in Section 5,

$$\mathbf{w}^\top \mathbf{s}_v \le \frac{\beta_{\text{upd}}}{\delta(1-\gamma)} \le H_{\max}.$$

The projection is therefore inactive, and

$$h_v = s_{v,1}.$$

Substituting $R = 1$ and $g_{v,k} = 1$ into Eq. (4) gives Eq. (6). □

Proof of Proposition 4.6. By Proposition 4.4, the state after the n-th event is

$$h_v^{(n)} = \beta_{\text{upd}} \tanh(\alpha p) \sum_{j=0}^{n-1} e^{-\mu_1 j / r_v}.$$

This is a geometric series with ratio

$$e^{-\mu_1 / r_v} < 1.$$

Its limit is Eq. (7). Using

$$1 - e^{-x} = x + o(x), \qquad x \to 0,$$

gives the asymptotic rate. □

Proof of Theorem 5.1. By Definition 3.3, $\Delta_v(t) \in [0,1]$, and

$$|\tanh(x)| \le 1.$$

Each update therefore satisfies

$$|h_v(t^+)| \le \gamma\, |h_v(t^-)| + \beta_{\text{upd}}.$$

Unrolling the recursion from $h_v = 0$ yields

$$\begin{aligned} |h_v(t)| &\le \sum_{i=0}^{\infty} \beta_{\text{upd}} \gamma^i \\ &= \frac{\beta_{\text{upd}}}{1-\gamma}. \end{aligned}$$

□

Proof of Corollary 5.2. Since $\log(1+\lambda_v) \ge 0$, the gain satisfies

$$\eta_v(t) \le \frac{\beta_{\text{upd}}}{\delta}.$$

The proof of Theorem 5.1 then applies with β_{upd} replaced by $\beta_{\text{upd}}/\delta$. □

Proof of Proposition 5.3. The difference between the two states is

$$d_v = h_v - h'_v.$$

Because both runs use the same gain η_v, it evolves as

$$\begin{aligned} d_v \leftarrow{}& \gamma\, d_v \\ &+ \eta_v \big(\tanh(\alpha \Delta_v) - \tanh(\alpha \Delta'_v)\big). \end{aligned}$$

Since tanh is 1-Lipschitz and

$$\eta_v \le \frac{\beta_{\text{upd}}}{\delta},$$

each increment has magnitude at most

$$\frac{\alpha \beta_{\text{upd}} \varepsilon}{\delta}.$$

Unrolling the recursion from $d_v = 0$, as in Theorem 5.1, yields Eq. (17). □

Proof of Corollary 5.4. At event e_j, the two states diverge by at most

$$\begin{aligned} &\eta_v \left|\tanh(\alpha \Delta_v) - \tanh(\alpha \Delta'_v)\right| \\ &\le \frac{\beta_{\text{upd}}}{\delta}. \end{aligned}$$

Every later event adds the same increment to both states and multiplies their difference by γ. □

Proof of Proposition 5.5. Unrolling Eq. (6) from $h_v = 0$ gives

$$h_v = \sum_{j=1}^{n} \gamma^{n-j} \eta(j)\, \phi(\Delta_v^{(j)}),$$

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where $\Delta_v^{(j)}$ is the evidence of the j-th event of v. For every j,

$$\mathbb{E}[\phi(\Delta_v^{(j)})] = \phi(p_0) + q_v\big(\phi(p_1) - \phi(p_0)\big).$$

Linearity of expectation yields Eq. (18). Part (i) follows because

$$G_n > 0, \qquad \phi(p_1) > \phi(p_0),$$

as tanh is strictly increasing and $p_1 > p_0$.

For part (ii), the gain $\eta(j)$ is nonincreasing in j, so

$$\eta(j) \geq \eta(n), \qquad j \leq n.$$

Moreover,

$$\sum_{j=1}^{n} \gamma^{n-j} = \frac{1-\gamma^n}{1-\gamma},$$

which gives the lower bound.

For the upper bound, split the sum at $j = n/2$. For $j > n/2$,

$$\eta(j) \leq \frac{\beta_{\text{upd}}}{\log(1+n/2)+\delta},$$

and the corresponding weights sum to at most

$$\frac{1}{1-\gamma}.$$

For $j \leq n/2$,

$$\eta(j) \leq \frac{\beta_{\text{upd}}}{\delta}.$$

Their weights form a geometric tail starting at an exponent of at least $n/2$, and hence sum to at most

$$\frac{\gamma^{n/2}}{1-\gamma}.$$

Both bounds are $\Theta(1/\log n)$. □

Proof of Proposition 5.6. In the fluid model, the background accumulation is

$$\int_0^\infty b\,\kappa(x)\,dx,$$

while the excess due to the anomaly is

$$\int_\tau^{\tau+L} c\,\kappa(x)\,dx.$$

For $\kappa(x) = e^{-\mu x}$, these equal

$$\frac{b}{\mu} \qquad \text{and} \qquad \frac{c}{\mu}e^{-\mu\tau}(1-e^{-\mu L}),$$

respectively, whose ratio is Eq. (20).

For part (i), differentiating

$$(1-e^{-\mu L})e^{-\mu\tau}$$

with respect to μ gives a stationary point when

$$L\,e^{-\mu L} = \tau(1-e^{-\mu L}),$$

or, equivalently,

$$e^{\mu L} = 1 + \frac{L}{\tau}.$$

The function is positive on $(0,\infty)$ and tends to zero at both ends, so this stationary point is its maximum.

For part (ii), the background and excess of the mixture are sums of component terms weighted by $w_i b_i$. Dividing the total excess by the total background expresses C_κ as

$$C_\kappa = \sum_i \pi_i C_{\mu_i},$$

where

$$\pi_i = \frac{w_i b_i/\mu_i}{\sum_j w_j b_j/\mu_j}.$$

The weights form a convex combination, and each term is nonnegative. □

Reading the defaults. The bounds translate the hyperparameters into interpretable quantities. With $\gamma = 0.95$, the evidence of an event halves after

$$\frac{\ln 2}{-\ln\gamma} \approx 13.5$$

further events of the same node. With $\beta_{\text{upd}} = \delta = 1$, Corollary 5.2 confines every state to

$$h_v \in [0, 20].$$

Any $H_{\max} \geq 20$ therefore leaves the projection of Definition 4.3 inactive in the scalar regime of Corollary 4.5.

REFERENCES